CONCEPT NOTE

AI LEARNING TUTOR

An AI-powered personalized learning assistant for students

Selected SDG	SDG 4 — Quality Education
Project Type	Educational technology / AI-enabled web application
Proposed Users	School students, college learners, educators and learning centres
Project Stage	Concept proposal; features and impact measures to be validated through a pilot

1. Project Summary

AI Learning Tutor is a proposed web-based learning assistant that helps students understand topics, practise questions and monitor their progress. A learner selects a subject or topic, chooses an explanation level, reviews a short explanation, and completes practice questions. The system uses quiz responses and learner feedback to recommend topics for revision. The project is designed as a support tool for learning—not as a replacement for teachers, textbooks or formal assessment.

2. Sustainable Development Goal (SDG)

Selected SDG: SDG 4 — Quality Education. The reference PDF presents SDG 4 as an opportunity for AI projects such as personalized learning agents, regional-language learning support and quizzes based on student performance. This concept follows that direction by proposing accessible explanations, adaptive practice and progress-based recommendations.

The intended contribution is to make additional learning support easier to access and more responsive to different learning needs. The project will assess its contribution through participation, completion and learning-improvement indicators during a small pilot.

3. Problem Statement

Students often learn at different speeds and have different levels of prior knowledge. In a busy classroom or during independent study, a learner may not receive immediate clarification or enough practice on a difficult topic. General online resources can be useful, but learners may need to search across several sources and may not know which concept to revise next.

Problem statement: Students need affordable, easy-to-use and personalized learning support that explains concepts clearly, provides guided practice and helps identify topics that require further revision.

4. Proposed Solution

Develop a responsive web application that combines a conversational learning interface with short topic explanations, practice quizzes and a progress dashboard. The first version should focus on a limited set of subjects and topics so that content quality and usefulness can be evaluated before expanding.

- **Topic-based explanations:** learners select a topic and receive a structured explanation with examples.
- **Practice quizzes:** short questions test understanding and provide immediate feedback.
- **Personalized revision:** quiz outcomes suggest topics to review again.
- **Progress dashboard:** learners can view completed activities, scores and progress over time.
- **Accessibility:** a mobile-friendly interface, simple language and readable layouts; regional-language support can be explored after the first version.
- **Educator review:** teachers or subject reviewers can check selected learning materials and flag errors.

5. Objectives

- Create a working prototype of an AI-supported personalized learning platform.
- Help learners identify knowledge gaps through topic quizzes and feedback.
- Offer clear explanations and examples matched to the learner's selected level.
- Track usage and learning indicators to evaluate whether the prototype is helpful.
- Align the project with SDG 4 by exploring a practical way to improve access to learning support.

6. Target Users and Beneficiaries

- **Primary users:** school students and college learners who want extra explanations and practice.
- **Secondary users:** teachers, tutors and learning centres that want to direct learners toward revision topics.

- **Potential institutional partners:** schools, colleges, libraries and education-focused community organisations.

The initial pilot should recruit a small, voluntary group of learners and educators. Their feedback will help identify confusing explanations, missing topics, accessibility issues and features that are most useful.

7. Key Features and User Flow

Suggested user flow: Sign in or continue to demo → choose subject and topic → select explanation level → read explanation and examples → complete quiz → review feedback → receive revision suggestions → check progress dashboard.

- Learner profile and selected learning goals.
- Topic library with reviewed content.
- AI-generated or AI-assisted explanations with clear limitations and review options.
- Quiz engine with scoring and answer explanations.
- Progress tracking and recommendations based on recent performance.
- Feedback/report option for incorrect or unclear content.

8. Implementation Approach

- **Phase 1 — Discovery:** interview learners and educators; select one or two subjects and define the first topic set.
- **Phase 2 — Prototype:** design the user interface and build the topic library, quiz flow and dashboard.
- **Phase 3 — AI integration:** connect an AI service for explanations and use prompt/content controls to keep responses focused on learning.
- **Phase 4 — Pilot testing:** test with a small group, collect feedback and compare quiz results before and after using the tool.
- **Phase 5 — Improvement:** fix confusing content, improve accessibility and decide whether more subjects or language options are justified.

A possible implementation stack is React for the frontend, Node.js/Express for the backend, a database for user progress, and an AI API for explanations. This is a suggested technical approach, not a requirement from the reference PDF.

9. Expected Outcomes

- A functional prototype that demonstrates personalized explanations, quizzes and progress tracking.
- A small, reviewed set of learning materials and practice questions.
- Evidence about learner satisfaction, usability and changes in quiz performance during the pilot.
- A prioritized list of improvements based on learner and educator feedback.
- A practical project model linked to SDG 4: Quality Education.

10. Monitoring and Evaluation

The pilot should establish baseline values before making claims about impact. Suggested indicators include:

- Number of learners enrolled and weekly active learners.
- Percentage of learners completing a lesson and its quiz.
- Change in scores between a baseline quiz and a comparable follow-up quiz.
- Learner-reported clarity and usefulness, measured through a short feedback form.
- Number of content issues reported and resolved.

Results should be reported honestly, including sample size and limitations. A small pilot can show usability and early signals, but it cannot by itself prove broad educational impact.

11. Risks, Ethics and Mitigation

- **Incorrect AI output:** use a bounded topic scope, educator review, feedback reporting and clear reminders to verify important information.
- **Privacy:** collect only necessary data, explain what is stored, protect accounts and provide a way to remove pilot data.
- **Over-reliance:** encourage learners to attempt questions themselves and use the tool alongside teachers and trusted materials.
- **Unequal access:** design for mobile screens and lower-bandwidth conditions where practical.
- **Bias or inaccessible content:** test with varied learners and review language, examples and readability.

12. Lean Canvas Alignment

The accompanying Lean Canvas summarises the proposed problem, solution, unique value proposition, unfair advantage, customer segments, key metrics, high-level concept, channels, cost structure and revenue streams. These are initial hypotheses. They should be validated through interviews, a prototype and a pilot rather than treated as established facts.

13. Sustainability and Future Scope

The prototype can begin with a free basic experience and limited content. If testing demonstrates value, possible sustainability models include an affordable premium plan, institutional subscriptions or sponsored access. Any paid model should preserve a useful basic learning path and be tested for affordability. Future extensions may include regional-language explanations, educator dashboards and additional subjects.

14. Conclusion

AI Learning Tutor proposes a practical, testable educational technology project aligned with SDG 4. By combining explanations, quizzes and progress-based revision suggestions, it aims to make independent study more structured and responsive. The next step is to validate the problem with learners and educators, build a small prototype and evaluate its usefulness before expanding.

Reference Basis

Primary reference: supplied PDF, *IBM SkillsBuild — Masterclass 5: Creating Real Life Projects, Project Guidance and Creating a Lean Canvas*. The reference introduces the Lean Canvas format (page 3), asks for a Lean Canvas PDF and Concept Note (page 8), and gives SDG 4 project examples including personalized learning and performance-based quizzes (page 6).

Note: The project idea and implementation details in this concept note are a proposed application of the reference structure; they are not presented as content copied from the reference.